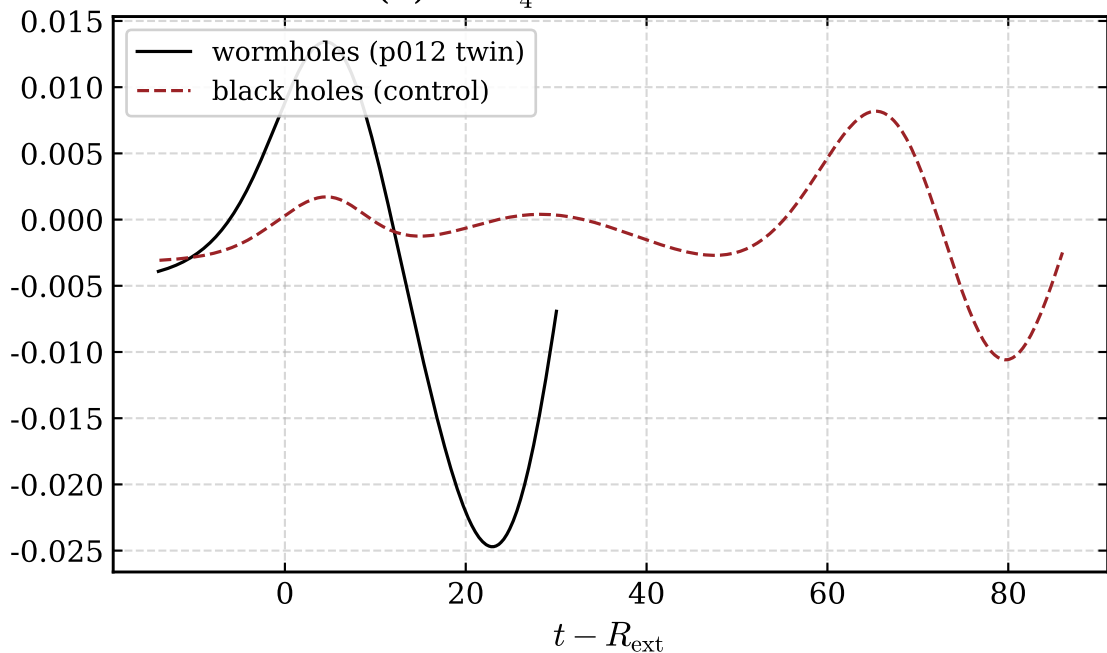
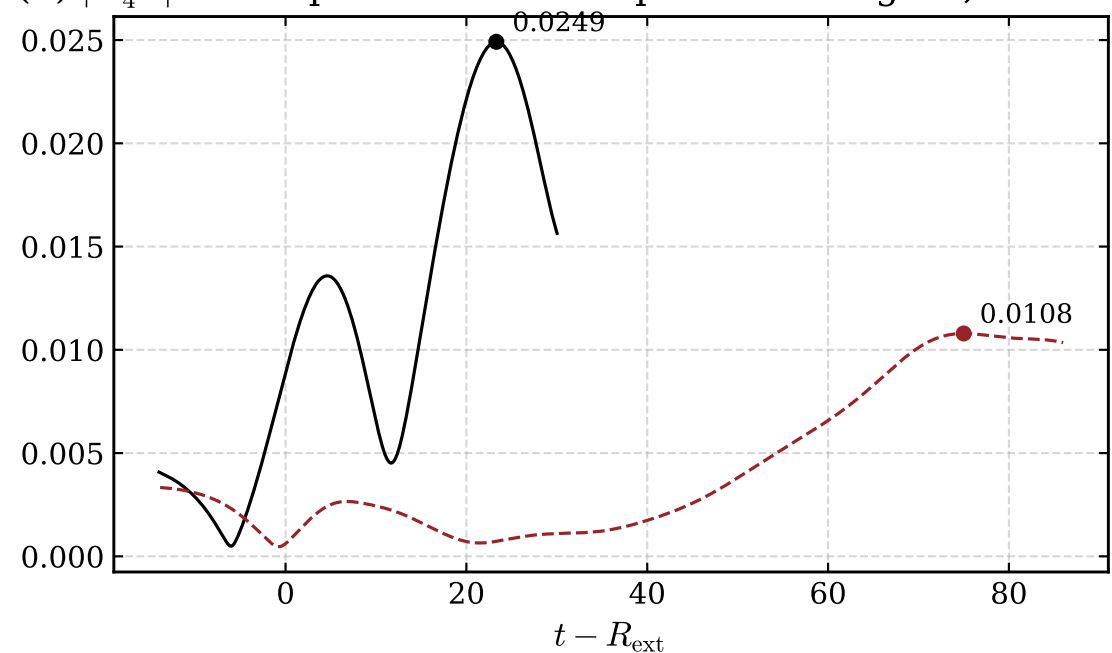


Same masses, same orbit ($d = 12$, $p = \pm 0.12$, $M_{\text{ADM}} = 1$ each) — different object

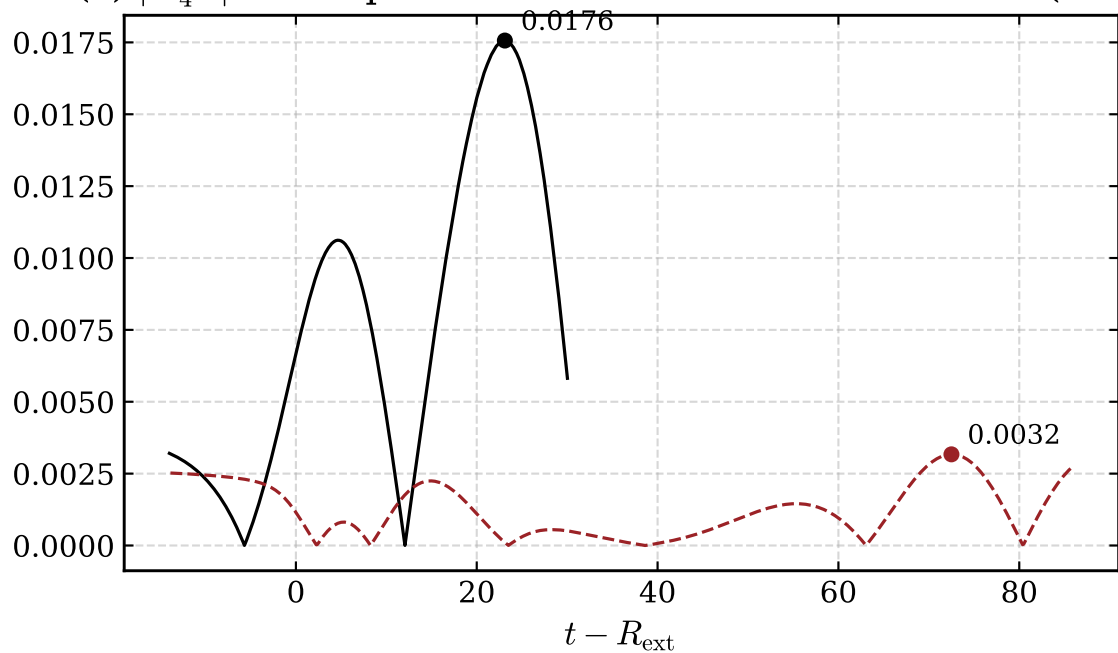
(a) $\text{Re } \Psi_4^{2,2}$ mode at $R = 14$



(b) $|\Psi_4^{2,2}|$ envelope — wormholes peak $2.3 \times$ higher, ~ 35 earlier



(c) $|\Psi_4^{2,0}|$ envelope — the wormholes' extra channel ($5.5 \times$)



(d) PSD of $\Psi_4^{2,2}$ — wormhole power sits lower in frequency

